MIDTERM EXAM
(Model B)

13. For the functions, n^k and c^n, what is the asymptotic relationship between these functions? Assume that k >= 1 and c> 1 are constants.
(a) n^k is Ω (c^n) (b) n^k is θ (c^n) (c) n^k is O(c^n) (d)None of mentioned

14. For the functions, f(n) and g(n) , what is the asymptotic relationship between these functions?

$$f(n)= \log_2 n^{\log_2 17}$$
$$g(n)= \log_2 17^{\log_2 n}$$

(a) f(n) = O (g(n)) (b) f(n) = Θ (g(n)) (c) f(n) = Ω(g(n)) (d)None of mentioned

15. Given the recurrence relation T(n)=2T(n/8)+$\sqrt[3]{n}$, determine which case of the Master Theorem applies and find the asymptotic complexity of T(n).
(a) Case1 (b) can't solve by master method (c) Case3 (d) Case2

16. Suppose you have a recurrence relation of the form T(n)=a·T(n/b)+f(n), where a=4, b=2, and f(n)=n^3. Which case of the Master Method applies, and what is the asymptotic complexity of T(n)?
(a) Case3: Θ(n^3) (b) Case2: Θ(n^2·logn) (c) Case 1: Θ(n^2) (d) None of mentioned

17. Give asymptotic tight bound for the recurrence T(n)= 0.5T(n/5) + 1
(a) θ(lg n) (b) Invalid function (c) θ(nlg n) (d) θ(n)

18. Given two square matrices, each with a size of n×n, what is the time complexity of the traditional matr multiplication algorithm? (a) O(n) (b)O(n^2) (c) O(n^3) (d) O(2^n)

19. What is the result of the recurrences which fall under second case of Master's theorem and f(n) =n^c
(a) T(n) = O(nlog$_b$a) (b) T(n)=O(f(n)) (c) T(n) = O(n^c logn) (d) T(n)=O(n^2)

20. This equation is .... $\sum_{k=1}^{n} 2^k = \Theta(2^n)$ (a) True (b) False

21. Recurrence equation formed for the tower of Hanoi problem is given by...
(a) T(n) = 2T(n-1)+n (b) T(n)=2T(n-1)+c (c) T(n)=2T(n/2)+n (d) T(n) = 2T(n/2)+c

22. The time complexity of Fibonacci is.... tower of Hanoi? (a) equal (b) smaller than (c) greater than

23. Choose the recursive formula for the Fibonacci series.(n>=1)
(a) F(n) = F(n-1) - F(n-2)+c (b) F(n) = F(n) + F(n+1)+c
(c) F(n) = F(n+1) + F(n+2)+c (d) F(n) = F(n-1) + F(n-2)+c

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Faculty of Computer Science and Information Technology
COM202: Algorithms
Fall 2024- 2025 & Time: 45 minutes
Dr. Salwa Osama & Dr Helal Ahmed

MIDTERM EXAM

(Model B)

Please read the following instructions carefully:

- Remember to fill in your Name and Student ID. Only sheets with both a name and ID will be graded.
- Deliver both external answer sheet and the question paper. If not, your paper will not graded.
- No Calculator is permitted.
- This is a closed-book exam. You are forbidden from accessing any external material, notes, electronic material, or online resources.

Name: _____ **ID:** _____

- Attempt to answer ALL questions in this exam.

Multiple choice Questions (Circle only ONE answer per question on the external answer sheet. If more than one answer is selected, the question will NOT be graded.) **[20 marks]**

1. The recurrence relation T(n)=T(n−1)+O(1) time complexity is ..... (a) O(n) (b)O(n^2) (c) logn (d) none
2. Consider the recurrence relation T(n)=5T(n/3)+n. What is the number of subproblems at level i of the recursion tree? (a) 3^i (b)5^{i-1} (c) 5^i (d) 5× 3^i
3. The recurrence relation T(n)=T(n−1)+O(1), can be solved by any recursive techniques? (a) True (b) False
4. The following statement is valid. 2^logn = Θ(n^2). (a) True (b) False
5. Given the recurrence T(n)=5T(n/3)+n, what is the complexity of the node at level i of the recursion tree?

   (a) $\frac{n}{3^i}$ (b) $\frac{5^i n}{3^i}$ (c) $5^i n$ (d) $\frac{5n}{3^i}$

6. The complexity for the following pseudocode is:

   i := n t := 0
   while (i ≤ n){ t := t + i; i := i+i}

   (a) Θ(lgn) (b) Θ(nlgn) (c) Θ(n√n) (d) Θ(C)

7. If all the elements in an input array is equal, What would be the running time of the Merge sort Algorithm?

   (a) O(nlogn) (b) O(n^2) (c) O(N) (d) None of the above

8. This equation cannot be solved using the master theorem T(n)=64*T(n/8) - nlogn? (a) True (b) Fals
9. Indicate the tightest bound of the following function f(N) = (N + N + N + N)^2

   (a) O(N^4) (b) O(N) (c) O(N^2) (d) O(1)

10. what is the best and the worst case of insertion sort are...

    (b) Ω (log N) , O(N) (b) Ω (1), O(N) (c) Ω (N^2), O(N) (d)None of them

    Ω (n), O(n^2)

11. Determine whether each of these functions is Θ(x), O(x^2) ,or Ω(x^2)

    f1=5^x f2=(x^3) * log(x^3) f3=log(x^x)

    (a) f1=Ω(x^2), f2=Ω(x^2), f3=O(x^2) (b) f1=Ω(x^2), f2=Ω(x^2), f3=Ω(x^2) (C)f1=Θ(x)
    (C) f2=Ω(x^2), f3=O(x^2) (d) a and b

12. What is the running time of the following Python functions in best case?

```
def fun1(B, x, y):
for i in range(B):
if x < y: for k in range(B * B): print("k =", k)
else: print("i =", i)
```

    (a) O(B ^2) (b) Ω(1) (c) Ω(B) (d)a and c

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